less dense than regions containing only rock, so their gravitational pull is weaker. This leads to a peculiar pattern of tidal gravitational forces.

In Figure 24.6, I use tendex lines to illustrate that tidal-force pattern. (See Chapter 4 for a discussion of tendex lines.) Squeezing tendex lines (drawn blue) stick out of the oil-bearing region, while stretching tendex lines (drawn red) stick out of the denser, oil-free region. As always, the two families of tendex lines are perpendicular to each other.

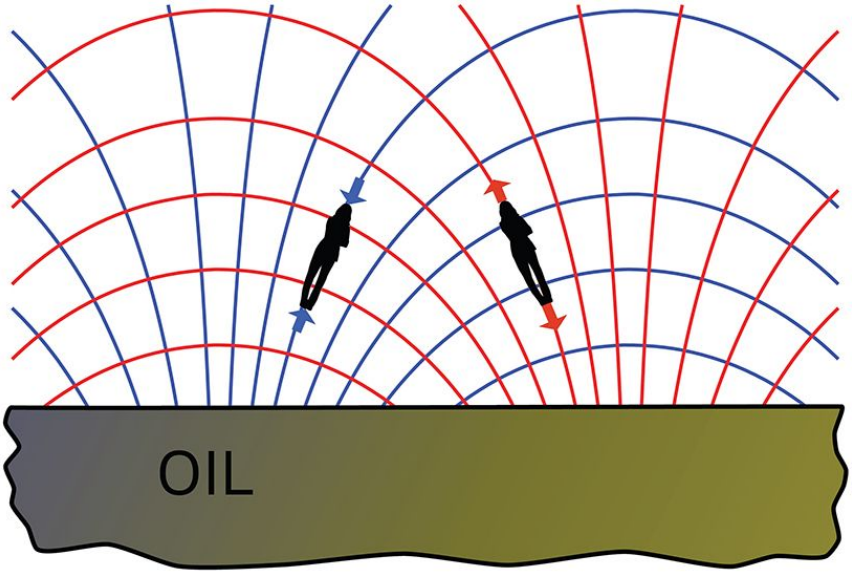


Fig. 24.6. Tendex lines above a portion of the Earth's crust. The red lines produce a tidal stretch along themselves. The blue lines produce a tidal squeeze.

An instrument called a gravity gradiometer can measure these tidal patterns (Figure 24.7). It consists of two crossed, solid rods attached to a torsional spring. On the ends of each rod are masses that feel gravity. The rods are normally perpendicular to each other, but in the figure the blue tendex lines squeeze the top two masses together and squeeze the bottom two together, while the red tendex lines stretch the right pair of masses apart and stretch the left pair apart. As a result, the angle between the rods decreases until the spring counterbalances the tidal forces. This is the gradiometer's readout, its "readout angle."